pharmaniaga®

ORS plus WHO recommended formulation since 2008

ORAL REHYDRATION SALTS

DESCRIPTION

A white and orange odour, crystalline powder.

COMPOSITION

Each sachet contains the equivalent of:

Sodium chloride	650 mg
Potassium chloride	375 mg
Trisodium citrate	725 mg
Glucose anhydrous	3.375 g

Ionic concentration (mmol/litre)

Na^+ 75, K^+ 20, Cl^- 65, Glucose 75 and Citrate 10

PHARMACODYNAMICS

Sodium chloride is the principal salt involved in maintaining the osmotic tension of the blood and tissues; changes in osmotic tension influence the movement of fluids and diffusion of salts in cellular tissues.

Solution containing sodium chloride are extensively used for the prevention or corrections of fluid and electrolyte deficits or imbalance in a wide range of clinical situations.

Sodium citrate, after absorption, is metabolized and has actions similar to those of sodium bicarbonate. It neutralizes the acid secretion in the stomach with the liberation of carbon dioxide. After absorption it is retained by the kidney to meet any deficit of bicarbonate in the plasma, e.g. in metabolic acidosis.

Potassium chloride is mainly used in the prevention and treatment of potassium deficiency. As it contains chloride ions it is employed in the treatment of hypokalaemia associated with hypochloraemic alkalosis.

Glucose is given by mouth as a readily absorbed carbohydrate in conditions associated with insufficiency of carbohydrates: it provides rapidly with available source of energy. It also reduces the need for the metabolism of fats and thus prevents ketonaemia.

PHARMACOKINETICS

Sodium chloride is readily absorbed from the gastro-intestinal tract. It is present in all body fluids but is mainly found in the extracellular fluid. The amount of sodium chloride normally lost in the sweat is small and the osmotic equilibrium is maintained by the excretion of surplus amounts in the urine.

Potassium salts other than the phosphate, sulphate, and tartrate are generally readily absorbed from the gastro-intestinal tract. Potassium is excreted mainly by the distal tubules of the kidney; 5 to 10 mmol a day maybe excreted in the faeces, and some in perspiration. The urinary excretion of potassium continues even when intake is low and faecal losses may be large in the presence of diarrhoea.

Glucose is absorbed from the gastro-intestinal tract. Three pathways of metabolism are established: glycolysis leading to the formations of pyruvate (aerobic) or lactate (anaerobic), followed by the Krebs tricarboxylic acid cycle (citric acid cycle), leading to metabolism to carbon dioxide and water, a pentose phosphate pathway leads also to carbon dioxide and water. Energy is released in these processes. Glucose is also stored as glycogen in the liver and muscle. Blood-glucose concentrations are maintained, on healthy persons, within normal limits by the insulin, which facilitates the passage of glucose through cell membranes, and other homeostatic mechanism. The body can metabolise about 800 mg per kg body-weight hourly. Glucose facilitates the absorption of sodium from the intestinal tract.

INDICATIONS

For replacement of water and electrolytes lost during moderate to severe diarrhoea.

CONTRAINDICATIONS

Glucose is contra-indicated in patients with the glucose-galactose malabsorption syndrome.

ADVERSE REACTIONS

Hypernatraemia is characteristic of excessive (sodium) administration including the overfeeding or inappropriate feeding of infants, in brain injury and acute water lack.

Symptoms of hypernatraemia may include restlessness, weakness, thirst, reduced salivation and lachrymation, swollen tongue, flushing of the skin, pyrexia, dizziness, headache, oliguria, hypotension, tachycardia, delirium, hyperpnea and respiratory arrest.

Retention of sodium and water – isotonic retention or iso-osmotic expansion because the plasma concentration in unchanged-leads to the accumulation of extra cellular fluid (oedema). This may effect the cerebral, pulmonary, or peripheral circulation.

The toxicity of potassium salts by mouth in healthy individuals is slight, since potassium is rapidly excreted in the urine. With some salts, such as potassium chloride, nausea vomiting, diarrhea and abdominal cramps may occur. Hyperkalaemia may occur after excessive intake (including the excessive transfusion of stored blood), after the use of potassium-sparing diuretics, in adrenal cortical insufficiency, renal failure, and acidosis and after tissue trauma. Symptoms include paraesthesia

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of the extremities, listlessness, mental confusion, weakness, paralysis, hypotension, cardiac arrhythmia, heart block, and cardiac arrest.

Concentrated glucose solutions given by mouth may cause nausea and vomiting.

WARNING AND PRECAUTIONS

Use with caution in renal impairment, severe dehydration, severe and continuing diarrhoea, glucose malabsorption, inability to drink, severe and continuing vomiting, intestinal obstruction, paralytic ileus or perforated bowel.

Any portion of the solution prepared using the oral powder that remains unused 24 hours after preparation should be discarded.

Do not boil. Do not add extra salt or sugar. Do not use if salt is wet. Consult physician if condition does not improve.

OVERDOSE AND TREATMENT

In the event of overdosage, hypernatraemia or hyperkalaemia could occur.

Hypernatraemia requires the use of sodium-free fluids and the cessation of excessive sodium intake. Very occasionally dialysis has been needed in severe hypernatraemia.

Iso-osmotic overload is managed by sodium and water restrictions plus measures to increase renal sodium and water loss such as 'loop diuretics' (e.g. frusemide) or, in specific circumstances, antimineralocorticoid agents.

Mild hyperkalaemia may be treated with sodium polystyrene sulphonate, administered by mouth or as an enema. In severe hyperkalaemia, treatment with haemodialysis or peritoneal dialysis may become necessary.

Hyperkalaemia associated with hyponatraemia may respond to treatment with infusions of sodium salts.

RECOMMENDED DOSAGE

Dissolve contents in each sachet of 5.145g in 250mL of cool boiled water. To be taken orally.

ROUTE OF ADMINISTRATION

Oral

STORAGE CONDITIONS

Store below 30°C.
Protect from light.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

Sachet of 5.145g

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Product Registration Holder/Manufacturer:
Pharmaniaga Manufacturing Berhad (198001006232)
No.11A, Jalan P/1, Kawasan Perusahaan Bangi,
43650 Bandar Baru Bangi, Selangor Darul Ehsan, Malaysia
PRP0108a.7 / BRP0108a.1 301221